Girls Build Bots — Junior App Development

Badge-aligned workshop guide for robotics-team-led outreach

Level: Juniors (Grades 4–5) · **Badge:** Coding for Good — App Development **Series:** Coding for Good (Coding Basics · Digital Game Design · App Development) **Format:** One 90–120 minute session · **Ratio:** 1 mentor per 4–5 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on paper templates and (optionally) free browser tools.

> **Troop Leaders:** This curriculum is designed to *align with* the Junior Coding for Good badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: coding *for good*

App developers start with a **user** and their problem — never with the code. This badge is the purest "for good" of the series: find someone's real problem, design the fix.

Vocabulary: user, problem statement, feature, interface, prototype, wireframe, click test, feedback.

App Development (90–120 minutes)

Badge theme: Learning how app developers go from a user's problem to an app design that solves it.

Materials

- Paper wireframe templates (phone outlines, 4–5 screens per team)
- Sticky notes, markers, scissors
- "User cards" for interviews (a busy teacher, a grandparent who forgets medicine times, a kid who loses library books, a coach organizing rides)
- Optional: tablets/laptops to examine 2–3 real apps' design choices

Activity 1 — Reverse-Engineer an App (15 min)

Open (or screenshot) a familiar app. Designer questions: **Who is the user? What problem does it solve? What's the very first thing you see, and why?** Notice deliberate choices — big buttons for important actions, the main thing front and center. Apps are *designed*, not accidental.

Activity 2 — Interview Your User (20 min)

Teams of 2–3 draw a **user card**. A mentor role-plays that user; Scouts interview them: “Walk me through your day. When does the problem happen? What have you tried?” Teams write a **problem statement**: *[User] needs a way to [goal] because [barrier]*. Developers who skip this step build apps nobody needs.

Activity 3 — Wireframe the App (40 min)

Teams design a paper prototype (a **wireframe**) of 4–5 screens: - Welcome screen (name + what it does, in the user’s words) - Main screen (the one button that solves the core problem — make it big!) - What happens after the tap (the payoff screen) - One extra feature screen — *only if the user interview justifies it*

Mentors circulate with the developer’s question: “Which screen solves the problem statement?” If a screen doesn’t serve the user, cut it — real developers cut features constantly.

Activity 4 — Click Test with the User (15 min) + Wrap (10 min)

The mentor-user returns. Teams run a **click test**: the user “taps” paper buttons; the team swaps screens by hand. Does the user get their problem solved in three taps or fewer? Teams make one fix based on what confused the user. Closing: each team presents their app in one sentence — “*[App name]* helps *[user]* to *[solve problem]*.”

> **Robot moment (optional):** Share for 60 seconds how the team decides what its driver dashboard needs — the driver is the user, and the interface shows exactly what matters mid-match, nothing more.

Completion Checklist — App Development

By the end of the session, to earn the badge, each Scout can: - [ ] Identify a real app’s user and the problem it solves - [ ] Interview a user and write a problem statement - [ ] Design a multi-screen wireframe where every screen serves the user - [ ] Run a click test and improve the design from user feedback

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Mentor Guide (for the robotics team)

- **You don’t need a robot for this.** Paper, markers, and one role-playing mentor per user card carry the session.
- **Play the user well.** Be specific and a little stubborn (“I don’t understand this button”) — realistic confusion teaches more than politeness.
- **Enforce user-first.** Every design debate resolves with one question: “what did the user say they need?”
- **Cutting features is winning.** Praise teams that remove a screen — that’s mature design judgment.
- **Three taps or fewer.** It’s a great concrete bar for “simple enough.”

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Junior Coding for Good badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.